

Take-home exam: December 18, 2013.

Course name: Network Optimization

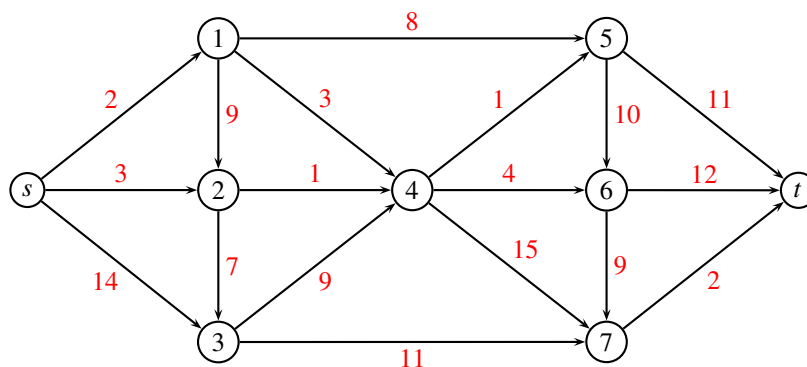
Course no: 42115

Rules: You are allowed to use your books, notes, slides, calculator, LP-solver etc. You are *not* allowed to get help from other people. You *must* fill in and sign the front page (last page in this document). You can write in Danish or English.

Assignment: The assignment consists of 17 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can print the figures, and use them in your answer by drawing on the figure.

Points: Small problems give 3 points, medium problems give 6 points, large problems 10 points, and xlarge problems give 20 points. A total of 100 points can be obtained.

Maximum Flow

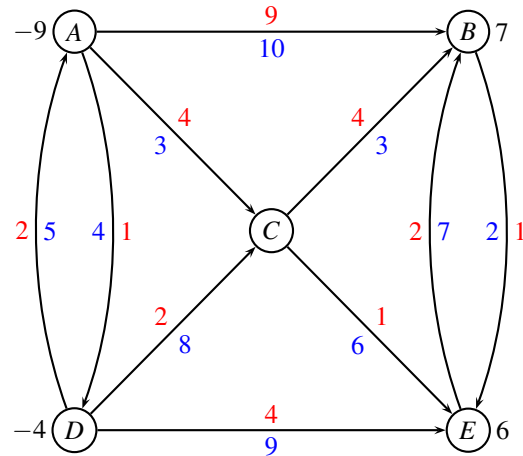


Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration write down which augmenting path was used, and how many units were sent. ■

Problem 2: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Problem 3: (small) Find first a legal start solution by solving an appropriate max-flow problem. ■

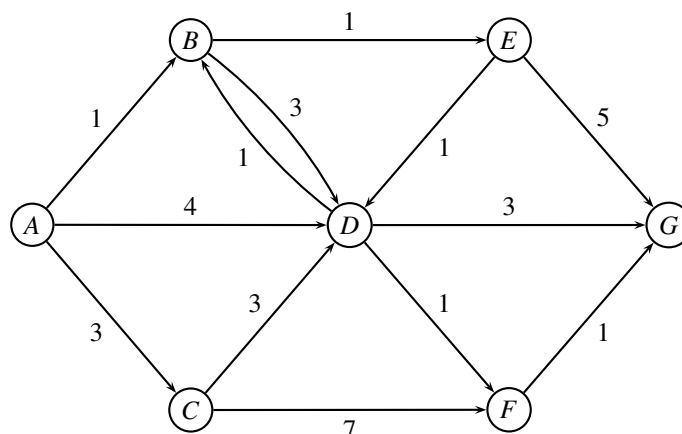
It is sufficient to draw the max-flow problem and to show the solution. No details about how the max-flow problem is solved are needed.

Problem 4: (large) Use the augmenting cycle algorithm on the above network. In each iteration, write down which augmenting cycle was used. ■

No details about how you found an augmenting cycle are needed.

Shortest path

Consider the following directed graph with distances w indicated on the edges:



Problem 5: (medium) Use Dijkstra's algorithm to solve the above shortest path problem starting at node A. Give sufficient details, so the algorithm can be followed. Draw the resulting shortest path tree. ■

We now consider a time constrained shortest path problem, where every edge counts as 1 time unit. So, for instance, the path $A \rightarrow D \rightarrow G$ has distance 2 since it passes two edges.

Problem 6: (small) Using the above shortest path tree, calculate the time consumption to every node. ■

We will now solve a time-constrained shortest path problem. For this purpose we use the Label Correcting algorithm from Slide 8 in the lecture “Resource Constrained Shortest Path Problem”.

Problem 7: (medium) Initiate each node with a list of labels $L_i = \{(w, t)\}$ containing one label (w, t) which is the distance and time found in question 6.

Use the Label Correcting algorithm to find the shortest path to every node using at most 4 time units. Give sufficient details so that your solution approach can be followed. Indicate the found solution tree and the found distances. ■

Nurse scheduling

A hospital administrator needs to establish a staffing schedule for nurses that will meet the minimum daily requirements shown in the following table

shift	1	2	3	4	5	6
time	6:00-10:00	10:00-14:00	14:00-18:00	18:00-22:00	22:00-2:00	2:00-6:00
min nurses	70	80	50	60	40	30

A work-plan for a nurse is called a duty. A duty normally consists of two consecutive shifts, although a duty consisting of only shift 6 is also allowed. A duty consisting of shift 6 followed by shift 1 is not allowed. The cost of a shift 1 to 4 is set to 1 unit. The cost of night shifts 5 and 6 is set to 2 units. The cost of a duty is the sum of shift-costs.

The administration wants to schedule the employees so that a sufficient number of nurses is available for each period while minimizing the overall costs.

Problem 8: (medium) Formulate the problem as a minimum cost flow problem (i.e. a graph with indicated edge costs, capacities, sources, and sinks). Hint: see Application 9.6 in Ahuja, Magnanti, Orlin, “Network Flows”. ■

Problem 9: (medium) Write up the network formulation from previous question as an LP-model ■

Problem 10: (small) Solve the model with CPLEX ■

Student assignment

In a academic program at DTU each master’s degree student is required to undertake a 6-month internship project at a company site. Since the projects are such an important component of the student’s educational program and vary considerably by company and by geography, each student would like to undertake a project of his or her liking. To assure that the project assignments are “fair”, the students and program administrators have decided to use an optimization approach: Each student ranks the available projects in order of increasing preference (1 indicate most preferred project). If the student

does not like a project, it is marked with a dash. The objective is to assign students to projects to achieve the best sum of total ranking of assigned projects. The project assignment has several constraints. Each student must work on exactly one project, and each project has an upper limit on the number of students it can accept. Each project must have a supervisor, drawn from a known pool of eligible faculty. Finally, each faculty member has an upper bounds on the number of projects that he or she can supervise. Also, each faculty member has preferences to which projects he or she wish to supervise (1 indicate most preferred project).

In a given semester we have 4 students, who have to be assigned to 3 projects. The preferences of the students are given in the following left table.

There are 4 faculty members for supervising the 3 projects. The preferences of projects are given in the right table. Notice, that all faculty members can supervise all projects.

student \ project	A	B	C
1	1	2	-
2	2	-	1
3	-	1	-
4	-	-	1
max students	2	1	2

project \ professor	U	V	W	X
A	1	1	3	2
B	2	2	2	3
C	3	3	1	1
max projects	2	1	2	1

Student preferences is the first priority, faculty member satisfaction second priority. Since all faculty members can supervise all projects, it is always possible to find a supervisor for a project. Every project needs to have a supervisor, but not all professors need to supervise a project. The supervisor uses the same resources for supervising a project with two students as a project with one student.

Problem 11: (large) Show how this problem can be solved using network flow techniques. (You should formulate it as two independant network problems. It is sufficient to draw the graphs and say which graph problem it is. You do not need to solve the graph problems). ■

Vehicle Routing Problem with time constraints

A delivery company has to visit 5 customers in some pre-described time-intervals $[a_i, b_i]$ as follows:

	earliest arrival	latest arrival
customer i	a_i	b_i
A	1	1
B	2	4
C	1	2
D	3	4
E	2	3

Each customer should be visited at least once.

The delivery company has 5 vehicles available, but not all vehicles need to be used if this is cheaper. All vehicles start at the depot s and return back to the same depot t (we use different names for start and end depot to make the modeling easier).

A *location* is either a depot or a customer. The cost of traveling between two locations is given in the following table (left). The next table (right) shows the travel time between two locations.

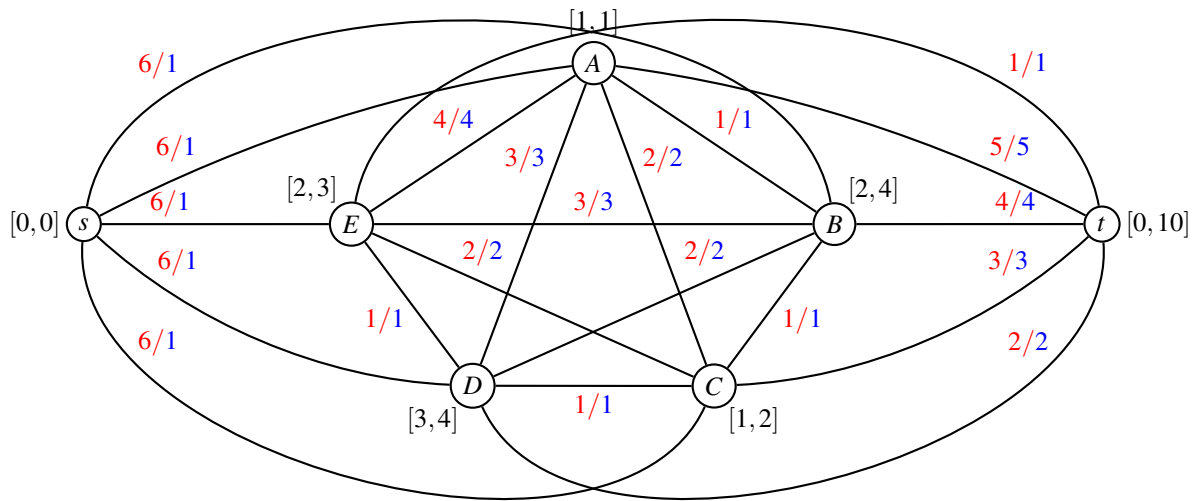
	cost						
	s	A	B	C	D	E	t
s	0	6	6	6	6	6	6
A	6	0	1	2	3	4	5
B	6	1	0	1	2	3	4
C	6	2	1	0	1	2	3
D	6	3	2	1	0	1	2
E	6	4	3	2	1	0	1
t	6	5	4	3	2	1	0

	time						
	s	A	B	C	D	E	t
s	0	1	1	1	1	1	1
A	1	0	1	2	3	4	5
B	1	1	0	1	2	3	4
C	1	2	1	0	1	2	3
D	1	3	2	1	0	1	2
E	1	4	3	2	1	0	1
t	1	5	4	3	2	1	0

It is assumed that the vehicle leaves depot s at time 0. Depot t has time window $[0, 10]$ meaning that all routes must be finished within 10 time units. If a vehicle arrives to a location before the specified time window, the vehicle will wait. Waiting does not impose any cost. All loading/unloading times at the locations are included in the driving times.

As an example, a route for a vehicle can start at the depot s at time 0. The vehicle first drives to customer C , arriving at time 1. The vehicle next drives to customer D , arriving at time 2. However, the time window of customer D is $[3, 4]$ so the vehicle waits one time unit until time 3 where it can service customer D . The vehicle then returns back to depot t , arriving at time 5. The cost of the route is $6 + 1 + 2 = 9$.

The problem can be represented by the following graph. Every location is represented by a node, and possible transitions between locations are represented by edges.



The red values indicate the costs, while the blue values indicate the travel time. The numbers $[a_i, b_i]$ indicate the time windows of node i .

Problem 12: (small) Find a feasible solution to the problem, using at most 4 vehicles. The route of each vehicle should satisfy the time windows. Report the cost of the solution. ■

We will use a path formulation of the problem, hence we start by writing up all possible routes for the vehicles. We have the following possible routes

route	nodes	cost
1	s,A,B,D,t	?
2	s,B,D,t	?
3	s,C,B,D,t	?
4	s,C,D,t	?
5	s,C,E,t	?
6	s,C,E,D,t	?
7	s,D,t	?
8	s,E,D,t	?
9	s,E,t	?

Dominated routes have been deleted. A route is dominated, if it is more expensive than another route and visits a subset of the nodes.

Problem 13: (small) Calculate the costs of all routes. ■

Problem 14: (small) Calculate the waiting time in each route. The waiting time occurs if the vehicle arrives at a customer before the time window. For instance the route s, C, D, t has a waiting time of 1 unit. ■

Problem 15: (medium) Write up an IP model with all routes and solve it to optimality using CPLEX. It is sufficient to consider undominated routes. ■

Since the number of possible routes can be very big for real-life problems, we wish to solve the problem through column generation. To start the process, we choose the trivial routes where each vehicle only visits one customer on the route:

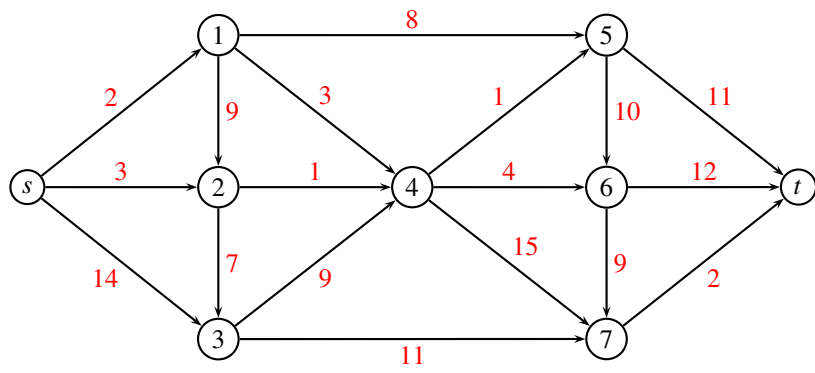
route	cost
s,A,t	11
s,B,t	10
s,C,t	9
s,D,t	8
s,E,t	7

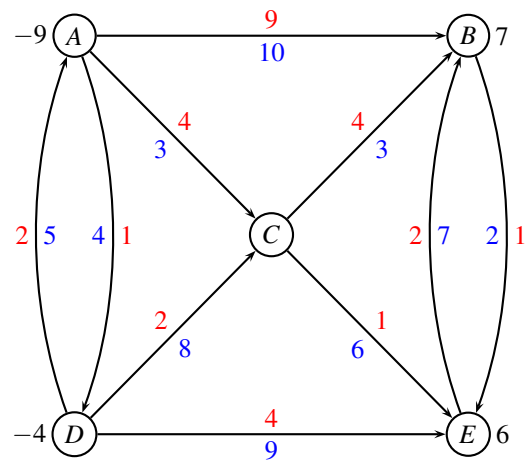
which gives a solution of 45.

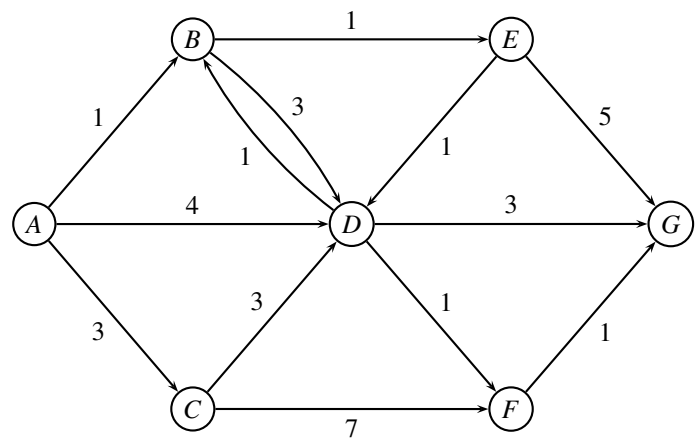
Problem 16: (xlarge) Use column generation to solve the problem to LP-optimality. In each iteration report dual values and show how the reduced costs are calculated. ■

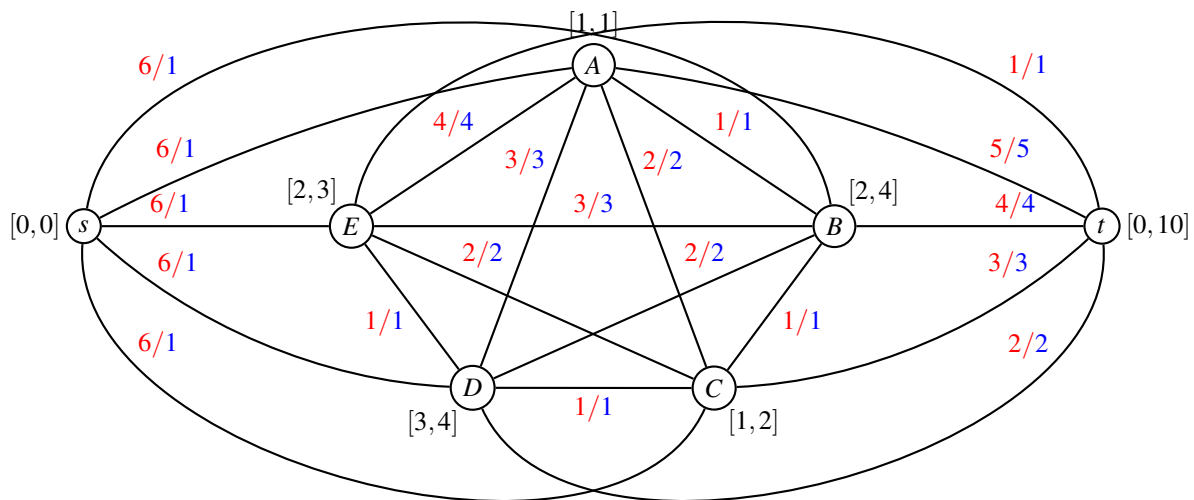
Problem 17: (small) Compare the LP solution with the integer solution, and discuss the quality of the bound. ■

end of assignment









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I hereby declare that I have worked alone on the project assignment and not discussed it with anybody else.

Date

Signature
